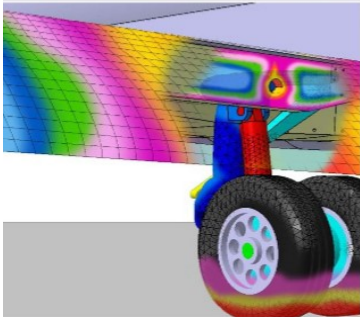


Qualification Pack



Airlines Ground Support Equipment Mechanic

Options: Maintain a safe and secure working environment

QP Code: AAS/Q0802

Version: 2.0

NSQF Level: 4

Aerospace and Aviation Sector Skill Council || No. 11, Dynamatic Park, Peenya
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Qualification Pack

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AAS/Q0802: Airlines Ground Support Equipment Mechanic

Brief Job Description

Individuals at this job are responsible for maintaining the mechanical systems of equipment and machinery. They use laid down procedures, and knowledge of the equipment to conduct routine maintenance and organize repairs. They are also involved in control and monitoring devices and occasionally in the manufacture of items that will help in maintenance

Personal Attributes

This job requires the individual to work independently and be judicious in making decisions pertaining to ones area of work. The individual must be physically fit to be able to maintain unusual working hours

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AAS/N0502: Follow safety and security procedures](#)
2. [ASC/N0001: Plan and organise work to meet expected outcomes](#)
3. [ASC/N1404: Carry out diagnosis of vehicle for repair requirements](#)
4. [ASC/N1405: Carry out service and major repairs in mechanical aggregates and overhauling of a vehicle](#)
5. [ASC/N1406: Carry out service and repairs of electrical and electronic faults in a vehicle](#)
6. [ASC/N1417: Carry out painting work on vehicles and replace/install painted body panels](#)
7. [ASC/N1419: Assist in the repair and replacement of vehicle body and chassis components](#)
8. [ASC/N9702: Carry out cleaning of a vehicle](#)
9. [ASC/N9708: Perform a routine check of basic functioning of vehicle and provide assistance to driver in servicing or minor repairs](#)
10. [ELE/N9905: Work effectively at the workplace](#)
11. [THC/N9916: Follow and Maintain Green Practices](#)

Options(Not mandatory):

Option : Maintain a safe and secure working environment

Perform Basic Life Support or basic first aid in medical emergency situations, as and when required.

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Respond to institutional emergencies appropriately.

1. [HSS/N9624: Maintain a safe and secure working environment](#)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
Sub-Sector	Maintenance Repair and Overhauling
Occupation	Maintenance, Maintenance
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7233.58
Minimum Educational Qualification & Experience	10th Class + I.T.I OR 12th Class/Diploma (Diploma in Mechanical / Automobile / Electrical engineering and allied branches) OR Not Applicable (Diploma ((Mechanical / Automobile / Electrical engineering) and allied branches)) with 2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	24/02/2022
Next Review Date	28/02/2026
NSQC Approval Date	24/02/2022
Version	2.0
Reference code on NQR	2022/AA/AASSC/05348
NQR Version	1.0

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AAS/N0502: Follow safety and security procedures

Description

This unit is about following awareness of safety and security procedures as defined by organisational policies and regulatory guidelines

Elements and Performance Criteria

Comprehending the safety and security procedures for conduct of operations

To be competent, the user/individual on the job must be able to:

- PC1.** comply with the organisations safety and security policies and procedures
- PC2.** comply with the regulatory guidelines on safe conduct of operations and maintenance of conditions to thwart any acts of unlawful interference
- PC3.** report any identified breaches of safety, and security policies and procedures to the designated person
- PC4.** coordinate with other resources at the workplace (within and outside the organisation) to achieve safe and secure environment
- PC5.** identify and mitigate any safety and security hazards like illness, accidents, fires or acts of unlawful interference if it falls within the limits of individuals authority
- PC6.** report any hazards outside the individuals authority to the relevant person in line with organisational procedures and regulatory guidelines
- PC7.** follow organisations emergency procedures for accidents, fires or acts of unlawful interference
- PC8.** identify and recommend opportunities for improving health, safety, and security to the designated person
- PC9.** complete all health and safety records are updates and procedures well defined

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** hazard identification and risk management as defined within the organisational policy and procedures
- KU2.** regulatory requirements and organisations procedures for maintenance of safety and security standards and individuals role and responsibilities in relation to this
- KU3.** how and when to report hazards
- KU4.** the limits of responsibility for dealing with hazards
- KU5.** the organisations emergency procedures for different emergency situations and the importance of following these
- KU6.** importance of maintaining high standards of safety and security
- KU7.** implications that any non-compliance with safety and security may have on individuals and the organisation
- KU8.** different types of breaches of safety and security and how and when to report these
- KU9.** evacuation procedures for workers and passengers

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- KU10.** how to summon medical assistance and the emergency services, where necessary
KU11. how to use the health, safety and accident reporting procedures and the importance of these
KU12. regulatory guidelines on dealing with safety and security emergencies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete accurately a well written report in English language detailing the situations of emergency with attention to detail
GS2. read instructions/guidelines/procedures/rules
GS3. listen to and orally communicate information with all concerned
GS4. make decisions on a suitable course of action or response if permitted by the authority matrix
GS5. monitor efficient functioning of all activities
GS6. plan and organize work to achieve targets and deadlines
GS7. communicate with passengers and other stakeholders in a courteous manner
GS8. maintain cordial work relationship
GS9. identify trends/common causes for errors and suggest possible solutions to the supervisor / management
GS10. identify and correct errors
GS11. analyse best possible solutions (cost, time, effort, etc.) suited for operations
GS12. concentrate on task at hand and complete it without error
GS13. apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Comprehending the safety and security procedures for conduct of operations</i>	52	48	-	-
PC1. comply with the organisations safety and security policies and procedures	5	5	-	-
PC2. comply with the regulatory guidelines on safe conduct of operations and maintenance of conditions to thwart any acts of unlawful interference	5	5	-	-
PC3. report any identified breaches of safety, and security policies and procedures to the designated person	5	5	-	-
PC4. coordinate with other resources at the workplace (within and outside the organisation) to achieve safe and secure environment	10	10	-	-
PC5. identify and mitigate any safety and security hazards like illness, accidents, fires or acts of unlawful interference if it falls within the limits of individuals authority	5	5	-	-
PC6. report any hazards outside the individuals authority to the relevant person in line with organisational procedures and regulatory guidelines	10	10	-	-
PC7. follow organisations emergency procedures for accidents, fires or acts of unlawful interference	2	3	-	-
PC8. identify and recommend opportunities for improving health, safety, and security to the designated person	8	2	-	-
PC9. complete all health and safety records are updates and procedures well defined	2	3	-	-
NOS Total	52	48	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N0502
NOS Name	Follow safety and security procedures
Sector	Aerospace and Aviation
Sub-Sector	Airline
Occupation	Loading & Unloading
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

ASC/N0001: Plan and organise work to meet expected outcomes

Description

This NOS unit is about planning and organising an individuals work in order to complete it to the required standards on time.

Elements and Performance Criteria

Work requirements including various activities within the given time and set quality standards

To be competent, the user/individual on the job must be able to:

- PC1.** keep immediate work area clean and tidy
- PC2.** treat confidential information as per the organisations guidelines
- PC3.** work in line with organisations policies and procedures
- PC4.** work within the limits of job role
- PC5.** obtain guidance from appropriate people, where necessary
- PC6.** ensure work meets the agreed requirements

Appropriate use of resources

To be competent, the user/individual on the job must be able to:

- PC7.** establish and agree on work requirements with appropriate people
- PC8.** manage time, materials and cost effectively
- PC9.** use resources in a responsible manner

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organisations policies, procedures and priorities for area of work, role and responsibilities in carrying out that work
- KU2.** the limits of responsibilities and when to involve others
- KU3.** specific work requirements and who these must be agreed with
- KU4.** the importance of having a tidy work area and how to do this
- KU5.** how to prioritize workload according to urgency and importance and the benefits of this
- KU6.** the organisations policies and procedures for dealing with confidential information and the importance of complying with these
- KU7.** the purpose of keeping others updated with the progress of work
- KU8.** who to obtain guidance from and the typical circumstances when this may be required
- KU9.** the purpose and value of being flexible and adapting work plans
- KU10.** how to complete tasks accurately by following standard procedures
- KU11.** technical resources needed for work and how to obtain and use these

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** write in at least one language
- GS2.** read instructions, guidelines/procedures
- GS3.** ask for clarification and advice from appropriate persons
- GS4.** communicate orally with colleagues
- GS5.** make a decision on a suitable course of action appropriate for accurately completing the task within resources
- GS6.** agree objectives and work requirements
- GS7.** plan and organise work to achieve targets and deadlines
- GS8.** deliver consistent and reliable service to customers
- GS9.** check own work and ensure it meets customer requirements
- GS10.** anomalies to the concerned persons
- GS11.** analyse problems and identify work-arounds taking help from
- GS12.** apply own judgement to identify solutions in different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work requirements including various activities within the given time and set quality standards</i>	16	47	-	-
PC1. keep immediate work area clean and tidy	2	9	-	-
PC2. treat confidential information as per the organisations guidelines	2	6	-	-
PC3. work in line with organisations policies and procedures	3	8	-	-
PC4. work within the limits of job role	3	6	-	-
PC5. obtain guidance from appropriate people, where necessary	3	7	-	-
PC6. ensure work meets the agreed requirements	3	11	-	-
<i>Appropriate use of resources</i>	9	28	-	-
PC7. establish and agree on work requirements with appropriate people	3	9	-	-
PC8. manage time, materials and cost effectively	3	11	-	-
PC9. use resources in a responsible manner	3	8	-	-
NOS Total	25	75	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N0001
NOS Name	Plan and organise work to meet expected outcomes
Sector	Automotive
Sub-Sector	Manufacturing and R&D, Sales and Service, Road Transportation
Occupation	Auto Components /Aggregates Repair
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

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ASC/N1404: Carry out diagnosis of vehicle for repair requirements

Description

This NOS unit is about troubleshooting problems and fault diagnosis of the vehicle (including both mechanical and electrical aggregates)

Elements and Performance Criteria

Carry the diagnosis of the vehicle identify the root cause of the trouble

To be competent, the user/individual on the job must be able to:

- PC1.** . understand the auto component manufacturer specifications related to the various components/ aggregates in the vehicle
- PC2..** understand the functioning of each system, component and aggregate (including both mechanical and electrical aggregates) of a vehicle
- PC3. .** follow standard operating procedures for using workshop tools and equipment for fault diagnosis or troubleshoot problem in a vehicle
- PC4. .** obtain sufficient information from customer/ service advisor to make an assessment of service and repair needs of the vehicle
- PC5..** review the job card and understand customer complaints
- PC6. .** follow standard operating procedure set out for diagnosing faults (in case of complex faults take the assistance of the senior diagnosis technician/ technical manager)
- PC7. .** use checklists and standard OEM operating procedures to understand if the fault is because of improper servicing, or low levels of oils, coolants, grease etc. or poor quality oil/ air filters etc.
- PC8..** conduct inspection of the engine and aggregates to diagnose need for repairs or adjustment in various engine aggregates
- PC9. .** conduct inspection of mechanical, electrical and electronic systems to diagnose need for repairs, adjustment or part replacement
- PC10..** dismantle and reassemble aggregates of a vehicle (with help from other technicians and helper)
- PC11. .** conduct routine and non-routine inspections for pre-purchase assessment, vehicle fitness assessment, emission testing, safety assessment, post-accident diagnostic assessment, post-repair serviceability assessment and manufacturer recall assessment
- PC12..** compare results of diagnostic inspections and tests against vehicle specifications and any regulatory requirements
- PC13. .** finalise the list all the service, repair and replacement requirements of the vehicle post the diagnosis in consultation with service advisor/ supervisor
- PC14. .** follow standard operating procedures for using workshop tools and equipment
- PC15. .** ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary
- PC16..** ensure any malfunctions observed in tools and equipment are reported to the concerned persons
- PC17..** understand the various precautions to be taken to avoid damage to the vehicle and its components while working on diagnosis or troubleshooting the vehicle for any faults

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- PC18..** request assistance from a superior when required(esp. in cases where the complex diagnosis needs to be done and is out of scope)
- PC19..** assist junior technicians in their work
- PC20. .** ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)
- PC21. .** drive a relevant 2/3/4 wheeler vehicle which is an important part of the diagnosis of the type of vehicle that is dealt by the relevant OEM

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard operating procedures of the organisation/ dealership for inspection and diagnosis of faults in a vehicle as prescribed by the OEM/ components manufacturer
- KU2.** standard operating procedures recommended by the dealership/ suppliers/OEM for using tools and equipment for diagnosis or troubleshooting of various aggregates
- KU3.** safety requirements for equipment and components during the diagnosis or troubleshooting the various aggregates for root cause analysis of the fault
- KU4.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer for the diagnosis of troubleshooting the vehicle for faults
- KU5.** organisational and professional code of ethics and standards of practice
- KU6.** safety, health, environmental policies and regulations for the workplace as well as for automotive trade in general (e.g. safe practices while working in pits/ under vehicles)
- KU7.** the basic technology used in and functioning of various components and aggregates of the vehicle including: engines and fuel system (diesel, petrol, electrical, gas, hybrid etc.) cooling system air supply systems emission and exhaust system ignition systems clutch assembly clutch operating system gearbox (manual and automatic) drivelines and hubs drive-train assembly and transmission systems (manual, automatic etc.) steering system suspension system brake system (including regenerative braking systems) tyres and wheels (including wheel alignment) radiator batteries and power storage system power-generating systems (including charging systems especially forelectrical and hybrid vehicles) electrical wire harness, lighting, ignition, electronic and air-conditioningsystems etc. energy recuperation systems, if applicable (e.g. in electric, gas and hybridvehicles) electronic systems including active and passive safety, media, comfort andconvenience, supplementary restraint systems (SRS), networking and othersystems electronic control unit hydraulic and pneumatic system various lubrication systems
- KU8.** the tools used to assess and confirm technical faults that cannot be determined through a visual inspection, including use of: organic light emitting displays anti-lock braking system abs/air bag scan tools, automotive scanners, graphing scanners, modular diagnostic information systems pressure indicators: fuel pressure testers, manifold gauge sets, oil pressure gauges, tire pressure gauges measuring equipment: vernier calipers, micrometer, feeler gauges, multimetre, flow metre, temp gauge, dial gauge etc. electrical and electronic testing equipment: volt meters, ammeters, ohmmeters, battery testing equipment, dedicated and computer based diagnostic equipment, oscilloscopes etc. other tools: laptops

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- KU9.** the various sources of information available for assessing service and repair requirements of the vehicle including: diagnostic displays visual inspections test drives vehicle/equipment manufacturer specifications standard operating procedures for diagnosis
- KU10.** typical symptoms of common technical faults in a vehicle including fluid levels, leaks, wear and tear, damage to a part/ aggregate and need for adjustments (e.g . HEOC- High Engine Oil Consumption, oil leakage, tyre wear etc,)
- KU11.** the various values and tolerance limits of various components across the mechanical/ electrical aggregates (e.g. within the engine assembly the following sub-aggregates : bore diameter, Liner fitment, piston height and butt clearance of piston rings, permissible imbalance in crankshaft (main and BE journal), axial and radial play in the camshaft etc.)
- KU12.** basic computer skills including the following: OEM specific computer applications basic computer based tasks (e.g. use of productivity tools such as word, excel etc.) basic internet based tasks (e.g. accessing and responding to emails etc.)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** create documentation required on the job (including diagnosis cards, work sheets, etc.) regarding the basic diagnosis and of various fault identification tests performed using various equipment as per the OEM/ auto component and aggregate guidelines
- GS2.** complete and maintain workplace records on inspection, diagnosis and repair activities
- GS3.** record all diagnostic inspections and tests carried out on a vehicle
- GS4.** write any additional work to be done (on the job card) basis the diagnosis of the vehicle (in major mechanical & electrical aggregates) and convey it to the service advisors and other technicians to carry out the work
- GS5.** write in at least one language
- GS6.** read and interpret workplace related documentation including job cards, safety instructions, OEM guidelines etc. from supervisors and the service advisor basis the information conveyed by the customer on the issue/ fault faced
- GS7.** read various sources of information available for vehicle diagnosis including service manual and diagnostic and visual displays put up in the workshop
- GS8.** read the technical brochures, new service schedules, service bulletins (refresher and new) released by the OEM/ auto components manufacturer related to diagnosis of the various faults across vehicle aggregates
- GS9.** read policies and regulations pertinent to the job, including OEM guidelines, health and safety instructions etc.
- GS10.** clearly communicate workplace information and ideas with workplace colleagues (verbal and non-verbal)
- GS11.** use terms, names, grades, and other nomenclature pertaining to the automotive trade, tools, specific workshop equipment etc.
- GS12.** communicate with colleagues to handle verbal enquiries, such as clarifying indicated faults and problems indicated on a job card which would lead to the proper diagnosis of the issue to do an effective root cause analysis

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- GS13.** communicate to the juniors technicians on the changes in technical brochures, new service schedules, service bulletins (refresher and new) released by the OEM/ auto components manufacturer related to diagnosis of the various faults across vehicle aggregates
- GS14.** communicate to the customer through the service advisor, the results of the test performed and appropriate values to find the root cause of the problem (e.g. in case of HEOC High Engine Oil Consumption issue, post the dismantled engine convey the appropriate condition of piston ring, crank shaft and cylinder block etc.)
- GS15.** analyse information and evaluate results to choose the best solution and solve problems
- GS16.** decide on the repair/ replacement of any aggregate (including those in the electrical and mechanical sub- assemblies) post the diagnosis (with help from a superior technical manager in case of complex diagnosis)
- GS17.** judge when to ask for help from a superior
- GS18.** plan work according to the required schedule and location
- GS19.** organize schedule to complete diagnosis on the vehicle so that repair/ replacement of aggregates/ components post diagnosis can start and vehicle can be delivered in a timely and cost effective manner
- GS20.** organise the workplace and work according to the principles of 5S
- GS21.** interpret the needs of customers understanding the key issue plaguing the poor performance of the vehicle and doing a proper diagnosis consulting the service advisor to minimise the repeat complaints
- GS22.** ensure that the service provided is of the highest order to ensure customer satisfaction
- GS23.** follow up with the Service Advisor on any unfavourable feedback received from customer on the complaints reported on the vehicle
- GS24.** recognise a workplace problem or a potential problem and take action prior to diagnosis (e.g. during diagnosis of the engine, ensure that engine aggregates are placed in proper place so that it doesn't cause any hindrance to other vehicles parked near the vehicle which is being diagnosed)
- GS25.** determine problems needing priority action
- GS26.** refer complex diagnosis outside area of responsibility to superior
- GS27.** analyse the complexity of work to determine if it can be successfully carried out (e.g. refer a vehicle to a superior in case of complex diagnosis for any fault which has not occurred before)
- GS28.** analyse the different components for the tolerance values post the diagnosis to understand the root cause of the issue
- GS29.** analyse, evaluate and apply the information gathered from observation, experience, reasoning, or communication to act efficiently
- GS30.** use the diagnosis results to take an appropriate decision on repair/ replacement of an aggregates (including mechanical and electrical subassemblies) in consultation with the supervisor/ service advisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry the diagnosis of the vehicle identify the root cause of the trouble</i>	30	70	-	-
PC1. . understand the auto component manufacturer specifications related to the various components/ aggregates in the vehicle	2	3	-	-
PC2.. understand the functioning of each system, component and aggregate (including both mechanical and electrical aggregates) of a vehicle	2	3	-	-
PC3. . follow standard operating procedures for using workshop tools and equipment for fault diagnosis or troubleshoot problem in a vehicle	1	2	-	-
PC4. . obtain sufficient information from customer/ service advisor to make an assessment of service and repair needs of the vehicle	1	2	-	-
PC5.. review the job card and understand customer complaints	1	2	-	-
PC6. . follow standard operating procedure set out for diagnosing faults (in case of complex faults take the assistance of the senior diagnosis technician/ technical manager)	1	2	-	-
PC7. . use checklists and standard OEM operating procedures to understand if the fault is because of improper servicing, or low levels of oils, coolants, grease etc. or poor quality oil/ air filters etc.	1	3	-	-
PC8.. conduct inspection of the engine and aggregates to diagnose need for repairs or adjustment in various engine aggregates	3	5	-	-
PC9. . conduct inspection of mechanical, electrical and electronic systems to diagnose need for repairs, adjustment or part replacement	3	6	-	-
PC10.. dismantle and reassemble aggregates of a vehicle (with help from other technicians and helper)	1	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. . conduct routine and non-routine inspections for pre-purchase assessment, vehicle fitness assessment, emission testing, safety assessment, post-accident diagnostic assessment, post-repair serviceability assessment and manufacturer recall assessment	3	6	-	-
PC12.. compare results of diagnostic inspections and tests against vehicle specifications and any regulatory requirements	2	6	-	-
PC13. . finalise the list all the service, repair and replacement requirements of the vehicle post the diagnosis in consultation with service advisor/ supervisor	1	5	-	-
PC14. . follow standard operating procedures for using workshop tools and equipment	1	3	-	-
PC15. . ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary	1	3	-	-
PC16.. ensure any malfunctions observed in tools and equipment are reported to the concerned persons	1	2	-	-
PC17.. understand the various precautions to be taken to avoid damage to the vehicle and its components while working on diagnosis or troubleshooting the vehicle for any faults	1	4	-	-
PC18.. request assistance from a superior when required(esp. in cases where the complex diagnosis needs to be done and is out of scope)	1	2	-	-
PC19.. assist junior technicians in their work	1	2	-	-
PC20. . ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)	1	2	-	-
PC21. . drive a relevant 2/3/4 wheeler vehicle which is an important part of the diagnosis of the type of vehicle that is dealt by the relevant OEM	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1404
NOS Name	Carry out diagnosis of vehicle for repair requirements
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N1405: Carry out service and major repairs in mechanical aggregates and overhauling of a vehicle

Description

This NOS unit is about an individual carrying out repairs and overhauling of mechanical systems of a vehicle.

Elements and Performance Criteria

Service and major repairs in mechanical aggregates

To be competent, the user/individual on the job must be able to:

- PC1..** understand the auto component manufacturer specifications related to the various components/ aggregates in the vehicle (including major aggregates like engine. gear box, transmission systems like propeller shaft etc.)
- PC2. .** follow standard operating procedures for using workshop tools and equipment for service and repairs of various vehicle aggregates
- PC3..** review the job card and understand service and repair work to be carried out in the various aggregates(including overall of various aggregates like engine)
- PC4. .** ensure that the correct spare parts, lubricants, tools and other materials required have been obtained
- PC5..** service, repair and overhaul: engines and fuel system (diesel, petrol, electrical, CNG, LPG etc.) cooling system radiator emission and exhaust system brake system clutch assembly gearbox, drive-train assembly and transmission systems (manual, automatic etc.) steering system suspension system tyres and wheels hydraulic and pneumatic system various lubrication systems
- PC6. .** conduct routine and non-routine inspections for vehicle fitness assessment, emission testing, safety assessment and post-repair serviceability assessment
- PC7. .** dismantle, assess, repair, clean, replace, adjust and reassemble the vehicle mechanical aggregates/ components
- PC8. .** identify and change vehicle components requiring change due to continuous wear and tear(including oil and air filters in the engine aggregate)
- PC9. .** ensure all dismantled components and aggregates are cleaned and conditioned prior to reassembly
- PC10..** carry out service, repair and overhauling activities safely to ensure: no damage to the vehicle or other vehicles no damage to vehicle components and systems no contact with hazardous materials
- PC11..** ensure disposal of materials (including waste oil, scrap of failed parts/ aggregates)in accordance with the organisations policies
- PC12. .** ensure, in consultation with the service advisor, approval of the customer on all repairs carried out
- PC13. .** record all service and repairs carried out and ensure completeness of tasks assigned before releasing vehicle for the next procedure

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- PC14.** . ensure any other repair requirements observed in the other components/ aggregates systems (like engine, gear box etc.) while repairing/ overhauling of braking systems are communicated to service advisor
- PC15.** . follow standard operating procedures for using workshop tools and equipment
- PC16.** . ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary
- PC17.** . ensure any malfunctions observed in tools and equipment are reported to the concerned persons
- PC18.** . inform the relevant persons where repairs/ overhauling of the aggregates are economically or technically infeasible
- PC19.** . request assistance from a senior technician when required
- PC20.** . assist junior technicians in their work
- PC21.** . ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard operating procedures for servicing, repair and replacement of parts in various aggregates of the vehicle
- KU2.** standard schedules and checklists recommended by the OEM for servicing and repair of various components and aggregates in the vehicle
- KU3.** safety requirements for equipment and components prescribed by the OEM (e.g. preventing/ dealing with oil spillage and inflammable materials)
- KU4.** identification codes, nomenclature of various components and aggregates
- KU5.** correct and appropriate grade of engine oil, coolant or lubricant grease to be used for the corresponding engine and other aggregates as specified by the OEM and auto component manufacturer
- KU6.** standard operating procedures recommended by the dealership/ suppliers/ OEM for using tools and equipment to be followed related to various aggregates and components in a vehicle as per the tool manufacturer instructions
- KU7.** standard operating procedures for rectification of errors in information (e.g. rectification of job card, reissue of correct tools and equipment etc.)
- KU8.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer
- KU9.** organisational and professional code of ethics and standards of practice
- KU10.** health and environmental policies and regulations for the workplace and automotive trade in general(e.g. safe working practices inside pits/ under vehicles)
- KU11.** regulatory requirements for vehicles including road safety, refrigerant handling, fuel storage and other requirements
- KU12.** operating specifications provided by the OEM for limits, fits and tolerances relating to engine mechanical, hydraulic and fluid systems for the vehicle

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- KU13.** the basic technology used in and functioning of various components and aggregates of the vehicle including: engines and fuel system (diesel, petrol, electrical, gas etc.) radiator emission and exhaust system brake system clutch assembly gearbox, drive-train assembly and transmission systems (manual, automatic etc.) steering system suspension system electrical wire harness, lighting, ignition, electronic and air-conditioning systems etc. electronic active and passive safety, media, comfort and convenience, supplementary restraint systems (SRS), networking and other systems electronic control unit tyres and wheels cooling system hydraulic and pneumatic system various lubrication systems
- KU14.** basic principles of: Ohms Law, voltage, power, current (AC/DC) resistance, magnetism, electromagnetism and electromagnetic induction etc. vehicle earthing and earthing methods vehicle engine systems (e.g. types, applications and operation of sensors, actuators, etc.) types of circuit protection and their use electrical safety procedures the operation of warning, charging and starter circuits symbols, units and terms associated with electric systems and components battery charging electrical/electronic control systems operation of electronic and electric engine systems (including electrical component function, electrical inputs, outputs, voltages and oscilloscope patterns, digital and fibre optics principles)
- KU15.** the tools used to assess and confirm technical faults that cannot be determined through a visual inspection, including use of: organic light emitting displays anti-lock braking system abs/air bag scan tools, automotive scanners, graphing scanners, modular diagnostic information systems pressure indicators: fuel pressure testers, manifold gauge sets, oil pressure gauges, tire pressure gauges pullers: ball joint separators, bearing pullers, gear puller tools, slide hammers specialty wrenches: alignment wrenches, chain wrenches, locking wrenches, lug wrenches trim or moulding tools: carbon scrapers, gasket scrapers, scrapers, spoons measuring equipment: vernier calipers, micrometer, feeler gauges, multimeter, flow metre, temp gauge, dial gauge etc. other tools: hand tools, power tools, lifting and jacking equipment, tensioning equipment, laptops, brake roller tester, chassis dynamometer, suspension activation, security activator etc. tools for other tasks such as cleaning of vehicles, tools, equipment and workshop
- KU16.** how to select the right materials for the job such as seals, sealants, fittings, gaskets, joints, fasteners etc.
- KU17.** how to carry out routine maintenance including: checking vehicle condition against OEM specifications to identify damage, corrosion, wear and tear, fluid levels, leaks and other problems in serviceability make adjustments to settings, alignment, pressures, tension, speeds and levels relevant to: - engine and aggregates (including fuel injection pump, ignition, intake and exhaust systems) - steering system - clutch and brake assembly - transmission system (including gearbox, differential, propeller shaft and axles) - other components (including tyres and body fittings)
- KU18.** the various sources of information available for assessing serviceability of the vehicle including: diagnostic displays visual inspections test drives vehicle/equipment manufacturer specifications standard operating procedures
- KU19.** how to repair and replace engine and its components including overhauling of the aggregates
- KU20.** how to dismantle, assess, repair, clean, condition, replace, adjust, reassemble and test mechanical components for correct operation
- KU21.** the method to correctly tag, seal and package checked engine components and aggregates
- KU22.** how to repair and replace engine and its components including overhauling of the aggregates

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- KU23.** how to dispose of replaced components in accordance with safety, health and environmental policies and regulations
- KU24.** how to measure/ inspect the machining or any other repair done from an outside source/ local machining garages on the various aggregates (like machining of the piston, cylinder head, cylinder block in the engine aggregate)
- KU25.** how to carry out the full overhaul of the engines (with support from the other technicians and machine shops from local garages)
- KU26.** precautions to be taken to ensure the following while working (including specific precautions to be taken when working with alternative fuel/ hybrid vehicles): no damage to the vehicle or other vehicles no damage to vehicle components and systems no contact with hazardous materials

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** create documentation required on the job (including job cards, work sheets, etc.) regarding the basic details of repair , maintenance and service performed on the various aggregates (including engine and other aggregates overhauling)
- GS2.** complete and maintain workplace records on inspection, diagnosis and repair activities
- GS3.** write any additional requirement of work on the vehicle for service, maintenance or repair found during the work done as specified in the job card (e.g. while working on the engine overhauling, if the clutch/ pressure plate is worn off, then convey to the superiors and service advisor)
- GS4.** write in at least one language
- GS5.** read job cards and instructions from service advisor regarding the repair and service of various vehicle aggregates
- GS6.** read various sources of information available for assessing service and repair requirements of vehicle aggregates including service manual and diagnostic and visual displays put up in the workshop
- GS7.** read policies and regulations pertinent to the job, including OEM guidelines, health and safety instructions etc. while working on the vehicle aggregates
- GS8.** read the technical brochures, new service schedules, service bulletins (refresher and new) released by the OEM/ auto components manufacturer related to service and maintenance of various vehicle aggregates
- GS9.** clearly communicate workplace information and ideas with workplace colleagues(verbal & non-verbal)
- GS10.** use terms, names, grades, and other nomenclature pertaining to the automotive trade, tools, specific workshop equipment etc.
- GS11.** communicate with colleagues and customers to handle verbal enquiries, such as clarifying instructions on a job card and responding to requests for information
- GS12.** interact with the customer through service advisor/ supervisor in case any additional repair and service needs to be done on the vehicle aggregates
- GS13.** communicate to the juniors technicians on the changes in technical brochures, new service schedules, service bulletins (refresher and new) released by the OEM/ auto components manufacturer related to service and maintenance of various vehicle aggregates

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- GS14.** analyse information and evaluate results to choose the best solution and solve problems
- GS15.** be proactive and creative in responding to workplace problems, changes and challenges
- GS16.** decide on the repair/ replacement of any engine aggregate post the diagnosis (with help from a technical manager in case required)
- GS17.** decide on whether the whole aggregate (gear box or engine overhaul) needs to be overhauled or some components/ parts needs to be changed
- GS18.** plan work according to the required schedule and location
- GS19.** plan the assigned aggregate repair work (including the specific engines or other aggregate that needs to be overhauled) on a daily basis and provide estimates of time required for each piece of work (e.g. by evaluating work assigned on a job card and providing time estimates for each service/ repair activity)
- GS20.** organise the workplace and work according to the principles of 5S
- GS21.** prioritise actions to achieve required outcomes
- GS22.** plan the assigned engine repair work (including the specific engines that needs to be overhauled) and provide estimates of time required for each engine overhaul repair
- GS23.** organise the schedule to complete work on the engine aggregate timely in case other aggregate repairs/ maintenance work is also required to be done
- GS24.** interpret the needs of customers by evaluating job cards and talking to Service Advisor and superiors
- GS25.** ensure that customer needs are assessed and satisfactory service is provided
- GS26.** follow up with the Service Advisor on any unfavourable feedback received from customer
- GS27.** recognise a workplace problem or a potential problem and take action
- GS28.** determine problems needing priority action
- GS29.** refer problems outside area of responsibility to appropriate person (e.g. unavailability of required spare parts or materials in the workshop)
- GS30.** gather information and provide assistance as required to solve problems
- GS31.** use a range of problem-solving techniques
- GS32.** create documentation required on the job (including job cards, work sheets, etc.) regarding the basic details of repair, maintenance and service performed on the various aggregates (including engine and other aggregates overhauling)
- GS33.** analyse the complexity of work to determine if they can successfully carry them out and if required, escalate to a superior
- GS34.** analyse, evaluate and apply the information gathered from observation, experience, reasoning, or communication to act efficiently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Service and major repairs in mechanical aggregates</i>	30	70	-	-
PC1.. understand the auto component manufacturer specifications related to the various components/ aggregates in the vehicle (including major aggregates like engine. gear box, transmission systems like propeller shaft etc.)	2	3	-	-
PC2. . follow standard operating procedures for using workshop tools and equipment for service and repairs of various vehicle aggregates	1	3	-	-
PC3.. review the job card and understand service and repair work to be carried out in the various aggregates(including overall of various aggregates like engine)	1	3	-	-
PC4. . ensure that the correct spare parts, lubricants, tools and other materials required have been obtained	1	3	-	-
PC5.. service, repair and overhaul: engines and fuel system (diesel, petrol, electrical, CNG, LPG etc.) cooling system radiator emission and exhaust system brake system clutch assembly gearbox, drive-train assembly and transmission systems (manual, automatic etc.) steering system suspension system tyres and wheels hydraulic and pneumatic system various lubrication systems	3	6	-	-
PC6. . conduct routine and non-routine inspections for vehicle fitness assessment, emission testing, safety assessment and post-repair serviceability assessment	3	5	-	-
PC7. . dismantle, assess, repair, clean, replace, adjust and reassemble the vehicle mechanical aggregates/ components	2	5	-	-
PC8. . identify and change vehicle components requiring change due to continuous wear and tear(including oil and air filters in the engine aggregate)	3	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. . ensure all dismantled components and aggregates are cleaned and conditioned prior to reassembly	1	4	-	-
PC10.. carry out service, repair and overhauling activities safely to ensure: no damage to the vehicle or other vehicles no damage to vehicle components and systems no contact with hazardous materials	2	5	-	-
PC11.. ensure disposal of materials (including waste oil, scrap of failed parts/ aggregates)in accordance with the organisations policies	1	3	-	-
PC12. . ensure, in consultation with the service advisor, approval of the customer on all repairs carried out	1	3	-	-
PC13. . record all service and repairs carried out and ensure completeness of tasks assigned before releasing vehicle for the next procedure	1	3	-	-
PC14. . ensure any other repair requirements observed in the other components/ aggregates systems (like engine, gear box etc.) while repairing/ overhauling of braking systems are communicated to service advisor	1	3	-	-
PC15. . follow standard operating procedures for using workshop tools and equipment	1	3	-	-
PC16. . ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary	1	3	-	-
PC17. . ensure any malfunctions observed in tools and equipment are reported to the concerned persons	1	2	-	-
PC18.. inform the relevant persons where repairs/ overhauling of the aggregates are economically or technically infeasible	1	2	-	-
PC19. . request assistance from a senior technician when required	1	2	-	-
PC20. . assist junior technicians in their work	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. . ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)	1	2	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1405
NOS Name	Carry out service and major repairs in mechanical aggregates and overhauling of a vehicle
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N1406: Carry out service and repairs of electrical and electronic faults in a vehicle

Description

This NOS unit is about an individual carrying out service and repairs within the electrical and electronic systems of a vehicle.

Elements and Performance Criteria

Service and repairs in electrical & electronic aggregates

To be competent, the user/individual on the job must be able to:

- PC1.** . understand the auto component manufacturer specifications related to the various electrical and electronic components and allied aggregates
- PC2..** follow standard operating procedures for using workshop tools and equipment for repair of electrical/ electronic components in a vehicle
- PC3. .** review the job card and understand work to be carried out in the electrical/ electronic aggregates as indicated by the supervisor or service advisor
- PC4. .** ensure that the correct spare parts tools and other materials required for service and repair of the electrical/ electronic components have been obtained

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- PC5. .** repair and overhaul: stability/steering/ suspension systems (including electronic stability systems, vehicle dynamic control, closed loop electronic steering and multi-class Bus systems) electric over hydraulic systems (including garbage compactors, crane rams, steering control, excavator bucket control, steering rudder control etc.) engine management systems (including fuel cell technology/hydrogen, on line maintenance and remote diagnostics, common rail diesel direct injection, drive by wire, multi-class Bus systems and closed loop diesel engine management systems) transmission/driveline systems (including clutches, torque converters, mechanical and automatic transmissions, drive and power take-off shafts and differentials, mechatronic modules and multi-class Bus systems) braking systems (including ABS, engine brakes, electric retarders, electric trailer brakes, brake by wire and multi-class Bus systems) safety systems (including fire suppressing, work load detecting, tyre pressure control, speed/load limiting, traction control, seat belt pre-tensioning, roll over protection, object detection, navigation aids, intelligent transport systems, intelligent SRS systems, adaptive cruise control, multi-class Bus systems, active and passive collision avoidance, infrared vision, lighting and windscreen wipers control) monitoring/protection systems (including display types such as LCD, VFD, CRT, HUD, re-configurable systems, electronic analogue display, on board diagnostics, remote/wireless monitoring systems and multi-class Bus systems) convenience and entertainment systems (including audio and visual units, compact disks, analogue tapes, radio, speaker types, amplifiers, crossovers, balancers, aerials and multi-class Bus systems) theft deterrent systems (including remote keyless entry (RKE), immobiliser system design, passive entry systems, two way RKE, fingerprint technologies, rolling codes, transmitter and receiver operation, satellite systems) electric and hybrid vehicle systems (including battery technology, motor drive systems, motor controllers, air conditioning systems, electronic protection systems and multi-class Bus systems) climate control systems (including air conditioning, heating, blending systems and multi-class Bus systems) gearbox, drive-train assembly and transmission systems (manual, automatic etc.) electrical wire harness, lighting, ignition, electronic and air-conditioning systems etc. electronic active and passive safety, media, comfort and convenience, supplementary restraint systems (SRS), networking and other systems electronic control unit hydraulic and pneumatic system
- PC6. .** PC6. repair all electrical and electronic faults including direct faults in: input sensors output actuators wiring harnesses computer systems calibration/adjustment specifications component specifications component assembly component damage system modifications
- PC7..** repair indirect faults caused on the major mechanical or other aggregates by the influence of electrical and electronic aggregate (e.g. influence of improper working on the ECU might have damaged the charging of the alternator
- PC8. .** remove, refit and test electrical components for normal operation following major/ minor body repair activities
- PC9..** dismantle, assess, repair, clean, replace, adjust and reassemble vehicle electric and electronic units
- PC10. .** ensure all dismantled components (other than the electrical or electronic components) are cleaned and conditioned prior to reassembly
- PC11. .** ensure disposal of materials (including scrap of failed parts/ aggregates) in accordance with the organisations policies
- PC12. .** understand the various precautions to be taken to avoid damage to other components/ aggregates of a vehicle while working on electrical/ electronic aggregates
- PC13. .** record all service and repairs carried out and ensure completeness of tasks assigned before releasing vehicle for the next procedure

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- PC14.** . ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary
- PC15..** ensure any malfunctions observed in tools and equipment are reported to the concerned persons
- PC16.** . request assistance from a senior technician or aggregate specialist when required
- PC17.** . inform the relevant persons where repairs are economically or technically infeasible
- PC18..** ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard operating procedures for servicing, repair and replacement of electrical/ electronic parts (including those related to various mechanical aggregates)
- KU2.** standard operating procedures recommended by the dealership/manufacturer/OEM for using tools and equipment for electrical/ electronic components
- KU3.** safety requirements for equipment within the tolerance limits used for service/ repair of electrical/ electronic components as prescribed by the OEM
- KU4.** identification codes, nomenclature of various electrical/ electronic components and aggregates
- KU5.** standard operating procedures for rectification of errors in information (e.g. rectification of job card, reissue of correct tools and equipment etc.)
- KU6.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer
- KU7.** organisational and professional code of ethics and standards of practice
- KU8.** safety, health and environmental policies and regulations for the workplace as well as for automotive trade in general(e.g. safe working practices inside pits/ under vehicles)
- KU9.** regulatory requirements for vehicles including road safety, refrigerant handling, fuel storage and other requirements
- KU10.** operating specifications provided by the OEM for limits, fits and tolerances relating to engine electrical, electronic and hydraulic and fluid systems for the vehicle
- KU11.** the basic technology used in and functioning of various components and aggregates of the vehicle including: engines and fuel system (diesel, petrol, electrical, gas etc.) radiator emission and exhaust system brake system clutch assembly gearbox, drive-train assembly and transmission systems (manual, automatic etc.) steering system suspension system electrical wire harness, lighting, ignition, electronic and air-conditioning systems etc. electronic active and passive safety, media, comfort and convenience, supplementary restraint systems (SRS), networking and other systems electronic control unit tyres and wheels cooling system hydraulic and pneumatic system various lubrication systems

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- KU12.** basic principles of: ohms Law, voltage, power, current (AC/DC) resistance, magnetism, electromagnetism and electromagnetic induction etc. vehicle earthing and earthing methods vehicle engine systems (e.g. types, applications and operation of sensors, actuators, etc.) types of circuit protection and their use electrical safety procedures the operation of warning, charging and starter circuits symbols, units and terms associated with electric systems and components battery charging electrical/electronic control systems operation of electronic and electric engine systems (including electrical component function, electrical inputs, outputs, voltages and oscilloscope patterns, digital and fiber optics principles) electrical theory and operation covering automotive digital computers, networked vehicles, voltage, current, resistance, power, capacitance, electrostatics, magnetics, inductance, discrete electronic components, logic families, and radio frequency
- KU13.** the tools used to assess and confirm technical faults that cannot be determined through a visual inspection, including use of: organic light emitting displays anti-lock braking system abs/air bag scan tools, automotive scanners, graphing scanners, modular diagnostic information systems pressure indicators: fuel pressure testers, manifold gauge sets, oil pressure gauges, tire pressure gauges pullers: ball joint separators, bearing pullers, gear puller tools, slide hammers specialty wrenches: alignment wrenches, chain wrenches, locking wrenches, lug wrenches trim or moulding tools: carbon scrapers, gasket scrapers, scrapers, spoons measuring equipment: vernier calipers, micrometer, feeler gauges, flow metre, temp gauge, dial gauge, analogue and digital multi-meters, lab oscilloscopes, data scanners, test lights, test LEDs, pulse generators etc. electrical and electronic testing equipment: volt meters, ammeters, ohmmeters, battery testing equipment, dedicated and computer based diagnostic equipment, oscilloscopes etc. other tools: hand tools, power tools, lifting and jacking equipment, tensioning equipment, laptops, brake roller tester, chassis dynamometer, suspension activation, security activator etc. tools for other tasks such as cleaning of vehicles, tools, equipment and workshop
- KU14.** how to modify and repair electric and electronic systems to correct faults including: varying the performance of DC motors to meet changes in operational requirements varying the performance of alternators to meet changes in operational requirements changing the electrical sequenced operating order of electric over hydraulic systems converting vehicle from ground to insulated return external modification (not within the computer) to a digital computer management system that enhances the system performance(e.g. modification to an electronic engine management system, improving the performance of an ECU controlled engine cooling fan system that necessitates changes to relay circuitry) external modification (not within the computer) to a digital computer management system, utilizing electronic circuit design, development, manufacture, trial, evaluation, improvement, and commissioning, that enhances the system performance (e.g. development of an electronic control unit to delay engine crank whilst sounding an alarm warning of impending start of hazardous equipment) internal modification (within the computer) to a digital computer management system, utilizing electronic circuit design, reprogramming, development, manufacture, trial, evaluation, improvement, and commissioning that enhances the system performance (e.g. rectifying an original internal computer design/operating deficiency, disabling a function no longer required by customer etc.)
- KU15.** . the various sources of information available for assessing serviceability of the vehicle including: diagnostic displays visual inspections vehicle/equipment manufacturer specifications standard operating procedures
- KU16.** how to dismantle, assess, repair, clean, condition, replace, adjust and reassemble and test electronic and electric components for correct operation

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- KU17.** the functioning of the vehicle battery and its schedule for change of water (as indicated by the battery manufacturer) and ensure that overcharging of the battery is avoided
- KU18.** how to dispose off replaced failed electrical/ electronic components in accordance with safety, health and environmental policies and regulations
- KU19.** precautions to be taken to ensure the following while working (including specific precautions to be taken when working with alternative fuel/ hybrid vehicles): no damage to the electrical / other advanced systems (in case of hybrid/ electrical vehicles) no damage to the vehicle on which work is being done along with other vehicles parked besides no damage to vehicle component sub-assemblies and other systems no contact with hazardous materials

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** create documentation required on the job (including job cards, work sheets, etc.) regarding the basic details of repair and maintenance done on the electrical/ electronic components
- GS2.** record all diagnostic performed on the electrical/ electronic components in vehicle
- GS3.** write in at least one language
- GS4.** write any additional requirement of work on the vehicle other than the one mentioned in the job card
- GS5.** read job cards and instructions from supervisors and the service advisor related to the work on the electrical/ electronic faults in a vehicle
- GS6.** read various sources of information available regarding the service and repair requirements of the electrical/ electronic sub-systems of the vehicle including service manual and diagnostic and visual displays put up in the workshop
- GS7.** read policies and regulations pertinent to the job, including OEM guidelines, health and safety instructions etc. related to work on the electrical/ electronic components and equipment
- GS8.** clearly communicate workplace information and ideas with colleagues (verbal & non-verbal)
- GS9.** use terms, names, grades and other nomenclature pertaining to the automotive trade, tools, specific workshop equipment etc.
- GS10.** communicate with colleagues and customers to handle verbal enquiries, such as clarifying instructions and responding to requests for information
- GS11.** interact with the customer through service advisor/ supervisor in case any additional work needs to be done related to the electrical/ electronic components which may not have been indicated in the job card
- GS12.** analyse information and evaluate results to choose the best solution and solve problems
- GS13.** decide on whether to repair or replace any electrical/ electronic aggregate post the diagnosis
- GS14.** judge when to ask for help from a colleague (Eq. regarding BS-3 engine, taking help from an engine specialist to solve the electrical issues related to the engine electronics)
- GS15.** plan work according to the required schedule and location
- GS16.** organise the schedule to complete work on the vehicle timely in case other aggregate repairs/ maintenance work is also required to be done
- GS17.** interpret the needs of customers by evaluating job cards and talking to service advisor and superiors

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- GS18.** ensure that the service provided is of the highest order to ensure higher levels of customer satisfaction
- GS19.** ensure timely communication of the additional requirements in a vehicle related to the electrical/ electronic components (including battery, headlight bulb change etc.) to the service advisor who in turn communicates it to the customer
- GS20.** follow up with the Service Advisor on any unfavourable feedback received from customer
- GS21.** recognise a workplace problem or a potential problem and take action (e.g. open wires while getting the battery charged)
- GS22.** determine problems needing priority action (e.g. any short circuit in any of the electrical circuit which may impact the performance of other aggregates esp. in a BS-3/ BS-4 vehicle which is entirely driven by electronic circuits)
- GS23.** refer problems outside area of responsibility to appropriate person (e.g. some defect in the ECU itself which would require special diagnosis by the senior technician/ supervisor)
- GS24.** gather information while working on electrical/ electronic aggregates and take appropriate action by consulting superiors (if needed)
- GS25.** assess repairs required based on technical faults identified as specified in the job card/ supervisor notes
- GS26.** refer complex problems (outside the current scope of work) to a superior in case any additional work requirement comes up
- GS27.** analyse, evaluate and apply the information gathered from observation, experience, reasoning, or communication to act efficiently
- GS28.** use the diagnosis results to take an appropriate decision on repair/ replacement of an electrical/ electronic aggregates

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Service and repairs in electrical & electronic aggregates</i>	25	75	-	-
PC1. . understand the auto component manufacturer specifications related to the various electrical and electronic components and allied aggregates	2	5	-	-
PC2.. follow standard operating procedures for using workshop tools and equipment for repair of electrical/ electronic components in a vehicle	1	5	-	-
PC3. . review the job card and understand work to be carried out in the electrical/ electronic aggregates as indicated by the supervisor or service advisor	1	5	-	-
PC4. . ensure that the correct spare parts tools and other materials required for service and repair of the electrical/ electronic components have been obtained	2	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. . repair and overhaul: stability/steering/ suspension systems (including electronic stability systems, vehicle dynamic control, closed loop electronic steering and multi-class Bus systems) electric over hydraulic systems (including garbage compactors, crane rams, steering control, excavator bucket control, steering rudder control etc.) engine management systems (including fuel cell technology/hydrogen, on line maintenance and remote diagnostics, common rail diesel direct injection, drive by wire, multi-class Bus systems and closed loop diesel engine management systems) transmission/driveline systems (including clutches, torque converters, mechanical and automatic transmissions, drive and power take-off shafts and differentials, mechatronic modules and multi-class Bus systems) braking systems (including ABS, engine brakes, electric retarders, electric trailer brakes, brake by wire and multi-class Bus systems) safety systems (including fire suppressing, work load detecting, tyre pressure control, speed/load limiting, traction control, seat belt pre-tensioning, roll over protection, object detection, navigation aids, intelligent transport systems, intelligent SRS systems, adaptive cruise control, multi-class Bus systems, active and passive collision avoidance, infrared vision, lighting and windscreen wipers control) monitoring/protection systems (including display types such as LCD, VFD, CRT, HUD, re-configurable systems, electronic analogue display, on board diagnostics, remote/wireless monitoring systems and multi-class Bus systems) convenience and entertainment systems (including audio and visual units, compact disks, analogue tapes, radio, speaker types, amplifiers, crossovers, balancers, aerials and multi-class Bus systems) theft deterrent systems (including remote keyless entry (RKE), immobiliser system design, passive entry systems, two way RKE, fingerprint technologies, rolling codes, transmitter and receiver operation, satellite systems) electric and hybrid vehicle systems (including battery technology, motor drive systems, motor controllers, air conditioning systems, electronic protection systems and multi-class Bus systems) climate control systems (including air conditioning, heating, blending systems and multi-class Bus systems) gearbox, drive-train assembly and transmission systems (manual, automatic etc.) electrical wire harness, lighting, ignition, electronic and air-conditioning systems etc. electronic active and passive safety, media, comfort and convenience, supplementary restraint systems (SRS), networking and other systems electronic control unit hydraulic and pneumatic system	2	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC6. . PC6. repair all electrical and electronic faults including direct faults in: input sensors output actuators wiring harnesses computer systems calibration/adjustment specifications component specifications component assembly component damage system modifications	2	6	-	-
PC7.. repair indirect faults caused on the major mechanical or other aggregates by the influence of electrical and electronic aggregate (e.g. influence of improper working on the ECU might have damaged the charging of the alternator	2	6	-	-
PC8. . remove, refit and test electrical components for normal operation following major/ minor body repair activities	2	5	-	-
PC9.. dismantle, assess, repair, clean, replace, adjust and reassemble vehicle electric and electronic units	2	5	-	-
PC10. . ensure all dismantled components (other than the electrical or electronic components) are cleaned and conditioned prior to reassembly	1	4	-	-
PC11. . ensure disposal of materials (including scrap of failed parts/ aggregates) in accordance with the organisations policies	1	4	-	-
PC12. . understand the various precautions to be taken to avoid damage to other components/ aggregates of a vehicle while working on electrical/ electronic aggregates	1	3	-	-
PC13. . record all service and repairs carried out and ensure completeness of tasks assigned before releasing vehicle for the next procedure	1	3	-	-
PC14. . ensure all workshop tools, equipment and workstations are adequately maintained by carrying out scheduled checks, calibration and timely repairs where necessary	1	4	-	-
PC15.. ensure any malfunctions observed in tools and equipment are reported to the concerned persons	1	3	-	-
PC16. . request assistance from a senior technician or aggregate specialist when required	1	2	-	-
PC17. . inform the relevant persons where repairs are economically or technically infeasible	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18.. ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)	1	2	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1406
NOS Name	Carry out service and repairs of electrical and electronic faults in a vehicle
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N1417: Carry out painting work on vehicles and replace/install painted body panels

Description

This OS unit is about the Painter carrying out painting work on vehicles and replace/ and install the painted body panels. It also includes carrying out specialised painting jobs in workshops which have specialised paint-shops within the body shop

Elements and Performance Criteria

Painting work on the parts and install the painted body panels

To be competent, the user/individual on the job must be able to:

- PC1.** . ensure that the vehicle is placed on a suitable platform, before the painting actually starts
- PC2.** . ensure that all the tools and equipment required is prepared, tested and adjusted, following manufacturers instructions prior to use, which includes: Painting equipment Paint Sanding and polishing tools Any other safety equipment
- PC3.** . ensure that the vehicle or body surface is free from any dust and dents prior to the actual painting
- PC4..** ensure that the dents (if found on the body surface) is repaired before starting to paint
- PC5.** . protect the vehicle and its contents when carrying out the repair of minor paint defects to paint surfaces
- PC6..** ensure that the sanding of the paint (ideally removing the paint to the bare metal, original primer) is done, sufficiently so that new paint adheres to it
- PC7.** . ensure that the surface is cleaned thoroughly, using mineral spirits or denatured alcohol to make sure that there is no oil (including body oils from fingers and hands) on the surface
- PC8.** . ensure that all surface not to be painted are covered with masking tape and paper (including glass, window trim, door handles, mirrors, grills) and ensure that there are no holes in the masking tape and paper to allow overspray to get through
- PC9.** . prime the surface with a corrosion resistant, self-etching primer after removing all paint down to bare metal by applying enough paint to fill scratches or pits left in the preparation process.
- PC10..** ensure that primer is cured thoroughly (as some primers might require recoating or applying fresh coat after a certain period)
- PC11.** . ensure that the surface is clean to remove any dust or oil that might have accumulated during priming
- PC12.** . ensure that prepared finished paint for spraying is sprayed on the vehicle/ body surface (using proper hardener/ catalyst in case the specific automotive enamel requires it)
- PC13.** . ensure the paint surface finish produced is free from dust and contamination between the beginning of the painting process and tack-free time
- PC14.** . polish the paint using a rubbing compound either by hand or through buffing machines/ power polishers especially on the edges
- PC15..** use suitable sources of technical information to support repair and replacement of vehicle parts

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- PC16.** . work in a way which minimises the risk of damage to the vehicle and other parts/ components
- PC17..** confirm that all the tools and equipment required are safe prior to use
- PC18.** . follow manufacturers instructions and correct procedures before replacing and installing vehicle parts
- PC19.** . correctly fit and balance the replaced and refitted parts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard operating procedures of the organisation/ dealership for inspection, and painting of vehicles or any other painted body panel
- KU2.** standard operating procedures for replacement of parts/ aggregates as mandated by the OEM which may be required to be dismantled while painting the vehicle or body panels
- KU3.** safety requirements for equipments (including sanding and polishing tools) and various paints and allied materials (primer, varnish, mineral oils, sand paper etc.) as prescribed by the OEM
- KU4.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified for carrying out paint related jobs
- KU5.** organisational and professional code of ethics and standards of practice
- KU6.** safety and health policies and regulations for the workplace as well as for automotive trade in general (e.g. safe practices while working in pits/ under vehicles)
- KU7.** the overall functioning of various components/ aggregates in a vehicle (as they may be affected while the surface is being painted)
- KU8.** basic technology used in the overall functioning of various types painting equipment
- KU9.** sanding techniques (along with filling & chiselling techniques) and various types of sandpaper to be used for a particular surface
- KU10.** various dust prevention techniques between the beginning of the painting process and tack-free time
- KU11.** various paints and their specific requirements including timing of the coat and the equipment/ tools used for each of the following: Primer (which acts as a leveller or protector and is the first coat to be applied) Solid Basecoat Paint (contains visual properties of colour and doesn't have any other effects) Metallic Basecoat Paint (contains aluminium flakes to create a sparkling and grainy effect called as metallic look) Clear Coat Paint (sprayed over the base paint to give a glossy or transparent coating)
- KU12.** process of auto spray painting
- KU13.** knowledge in mixing variation of colours and their effects on the painted surface
- KU14.** various refinishing methods which may enhance the look and feel of the painted/ coated surface
- KU15.** various rust prevention techniques in the long run

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** record and document the requirements of paint related jobs that needs to be performed on the vehicle
- GS2.** write in at least one language
- GS3.** read instructions mentioned on the job card given by Service Advisor/Supervisorsto the paint related jobs to be performed
- GS4.** interact with the Customer/ Service Advisor
- GS5.** communicate the damage caused to vehicle and its body parts
- GS6.** specify the corrective measures required to repair/replace the body component
- GS7.** interact with team members both in the workshop and the bodyshopto work efficiently
- GS8.** decide which body components are to be repaired and which needs to be replaced before starting the actual painting work
- GS9.** identify whether the surface is clean enough using mineral spirits or denatured alcohol to make sure that there is no oil (including body oils from fingers and hands)on the surface
- GS10.** identify the correct level of priming required and to what levels curing is required (as some primers might require recoating or applying fresh coat after a certain period)
- GS11.** identify type of paint (colour, texture etc.) to be used incorporating the customers preference
- GS12.** work according to required schedule and location
- GS13.** organise the schedule according to the vehicular population on which paint related work needs to be performed keeping the overall delivery date in mind
- GS14.** ensure that customer needs regarding the painting related jobsare assessed and satisfactory service is provided
- GS15.** suggest to customer incase of suitability of a better variety of paints/ polishing materials keeping the cost and durability factor in mind
- GS16.** inform the customer incase of time to be taken for the paint process to be complete with proper quality
- GS17.** inspect damaged vehicles and decide corrective action to be undertaken with respect to painting jobs
- GS18.** inspect equipment to ensure proper working order and take any corrective actions as required
- GS19.** inspect vehicles which do not conform to OEM standards (e.g. customised vehicles) and decide on a the correct method to be adopted for painting
- GS20.** check the usefulness of shop tools to see if they are suitable for work on new models of vehicles
- GS21.** evaluate the complexity of tasks to determine if they can successfully carry them out
- GS22.** evaluate the information gathered from the customer report/ job card and assess repairs

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Painting work on the parts and install the painted body panels</i>	20	80	-	-
PC1. . ensure that the vehicle is placed on a suitable platform, before the painting actually starts	1	4	-	-
PC2. . ensure that all the tools and equipment required is prepared, tested and adjusted, following manufacturers instructions prior to use, which includes: Painting equipment Paint Sanding and polishing tools Any other safety equipment	2	4	-	-
PC3. . ensure that the vehicle or body surface is free from any dust and dents prior to the actual painting	1	4	-	-
PC4.. ensure that the dents (if found on the body surface) is repaired before starting to paint	1	4	-	-
PC5. . protect the vehicle and its contents when carrying out the repair of minor paint defects to paint surfaces	1	4	-	-
PC6.. ensure that the sanding of the paint (ideally removing the paint to the bare metal, original primer) is done, sufficiently so that new paint adheres to it	1	5	-	-
PC7. . ensure that the surface is cleaned thoroughly, using mineral spirits or denatured alcohol to make sure that there is no oil (including body oils from fingers and hands) on the surface	1	4	-	-
PC8. . ensure that all surface not to be painted are covered with masking tape and paper (including glass, window trim, door handles, mirrors, grills) and ensure that there are no holes in the masking tape and paper to allow overspray to get through	1	4	-	-
PC9. . prime the surface with a corrosion resistant, self-etching primer after removing all paint down to bare metal by applying enough paint to fill scratches or pits left in the preparation process.	1	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10.. ensure that primer is cured thoroughly (as some primers might require recoating or applying fresh coat after a certain period)	1	4	-	-
PC11. . ensure that the surface is clean to remove any dust or oil that might have accumulated during priming	1	4	-	-
PC12. . ensure that prepared finished paint for spraying is sprayed on the vehicle/ body surface (using proper hardener/ catalyst in case the specific automotive enamel requires it)	1	5	-	-
PC13. . ensure the paint surface finish produced is free from dust and contamination between the beginning of the painting process and tack-free time	1	4	-	-
PC14. . polish the paint using a rubbing compound either by hand or through buffing machines/ power polishers especially on the edges	1	5	-	-
PC15.. use suitable sources of technical information to support repair and replacement of vehicle parts	1	4	-	-
PC16. . work in a way which minimises the risk of damage to the vehicle and other parts/ components	1	4	-	-
PC17.. confirm that all the tools and equipment required are safe prior to use	1	4	-	-
PC18. . follow manufacturers instructions and correct procedures before replacing and installing vehicle parts	1	4	-	-
PC19. . correctly fit and balance the replaced and refitted parts	1	4	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1417
NOS Name	Carry out painting work on vehicles and replace/install painted body panels
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N1419: Assist in the repair and replacement of vehicle body and chassis components

Description

This NOS unit is about an Auto-Body Technician who is responsible for assisting repair and replacement of the vehicle chassis and body components and identifies any other repairs requirements in the vehicle as a result of damage to the body / chassis components (including the chassis frame)

Elements and Performance Criteria

Assist in repair & replacement of vehicle body and chassis components

To be competent, the user/individual on the job must be able to:

- PC1.** . assist the senior body technician to assess the overall damage to vehicle body and chassis components
- PC2..** follow supervisors instructions as to which parts to restore or replace with respect to the auto body/ chassis components
- PC3.** . select, calibrate and use the appropriate tools and equipment for the body component/ chassis repair of the vehicle
- PC4.** . remove upholstery, accessories, electrical window-and-seat-operating equipment, and trim to gain access to vehicle bodies and fenders
- PC5.** . cut and tape plastic separating film to outside repair areas to avoid damaging surrounding surfaces during repair procedure, and remove tape and wash surfaces after repairs are complete
- PC6.** . assist in performing repair work under supervision of senior body technician such as: body panel repair minor structural damage major welded panels body components using lead wiping major sectional repair chassis/ frame and associated components
- PC7.** . assist in removal and replacement related work under supervision of Senior Body Technician such as: vehicle body panels, panel sections and ancillary fittings protector mouldings, transfers and decals mechanical units/assemblies electrical/electronic units/assemblies
- PC8..** install various vehicle component seals
- PC9.** . prepare surface and apply window tinting
- PC10.** . file, grind, sand and smooth fill or repair surfaces, using power tools and hand tools
- PC11..** assist in clamping frames and sections to alignment machines that use hydraulic pressure to align damaged components
- PC12.** . assist in removal of damaged sections of vehicles using metal-cutting guns, air grinders and wrenches, and install replacement parts using wrenches or welding equipment
- PC13.** . assist in mixing polyester resins and hardeners to restore damaged areas
- PC14.** . assist in fitting and securing windows, vinyl roofs, and metal trim to vehicle bodies, using caulking guns, adhesive brushes, and mallets
- PC15.** . assist in filling small dents that cannot be worked out with plastic or solder
- PC16.** . assist in removal of small pits and dimples in body metal using pick hammers and punches

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- PC17.** ensure any malfunctions or repair requirements observed in vehicles (and beyond own scope of work) are reported to the concerned person
- PC18.** ensure any malfunctions observed in tools and equipment are reported to the concerned person
- PC19..** clean work areas, using air hoses, to remove damaged material and discarded fiberglass strips used in repair procedures
- PC20..** ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard operating procedures of the Organisation/ Dealership for inspection and repair of vehicles for the body/ chassis components
- KU2.** standard operating procedures for repair and replacement of automotive body parts mandated by the OEM
- KU3.** safety requirements for equipment and components (e.g. preventing/ dealing with oil spillage and inflammable materials)
- KU4.** standard operating procedures recommended by the Dealership/ Suppliers/OEM for using tools and equipment to be followed as per the manufacturer instructions (e.g. maintaining safety while using welding and soldering equipment)
- KU5.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by the organization
- KU6.** organisational and professional code of ethics and standards of practice
- KU7.** safety, health, environmental policies and regulations for the workplace as well as for Automotive trade in general (e.g. safe practices while working in pits/ under vehicles)
- KU8.** workplace policies and schedules for housekeeping activities and equipment maintenance
- KU9.** the tools and methods used to assess and confirm structural damage that cannot be determined through a visual inspection, including use of a tape measure or atram bar, a tram gauge, and self-centering gauges
- KU10.** the kind of tools and procedures required based on the damage sustained by the vehicle including: power tools and hand tools (e.g. hammers) air hoses various clamps, holding jigs, relevant special equipment
- KU11.** drills, drill bits, bolts, nuts and washers
- KU12.** the storage location for the tools and materials used in the workshop
- KU13.** the procedure for measurement of : length squareness flatness angles roundness depth clearances any measurements that can be taken from analogue or digital devices
- KU14.** how to assist in repairing and fixing the following non-permanently attached body panels: wings doors bonnet boot lids & tail gates
- KU15.** the various welding techniques
- KU16.** The correct alignment of panels and components and the applicable methods
- KU17.** the various cleaning agents/sprays (de-waxing, detergents, degreasers, special purpose agents)
- KU18.** the appropriate tools required for structural repairs

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- KU19.** the repair procedure to be followed for particular damage
- KU20.** the measures to be adopted to protect the vehicle from damage before, during and after removing and fitting activities
- KU21.** appropriate personal and vehicle protection to be used for various procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and document the basic details of repairs performed on various body and chassis aggregates/ components
- GS2.** write in at least one language
- GS3.** read work order, specification, etc. related to the job including instructions mentioned on the job card given by service advisor/supervisors
- GS4.** read any specific safety related guideline (applicable for CNG/ LPG/ Electric vehicle)
- GS5.** communicate the damage caused to vehicle and its body parts
- GS6.** interact with the customer/ service advisor and senior technicians
- GS7.** interact with team members including colleagues in the workshop to work efficiently
- GS8.** judge when to seek assistance from a superior
- GS9.** plan work according to the required schedule and location
- GS10.** ensure that customer needs regarding the denting and body/ chassis related jobs are assessed and satisfactory service is provided
- GS11.** inspect damaged vehicles and assist in repairs required
- GS12.** inspect equipment to ensure proper working order and take any corrective actions as required
- GS13.** ensure that all dents which cant be repaired to be referred to superiors to take an appropriate decision
- GS14.** check the usefulness of shop tools to see if they are suitable for work on new models of vehicles
- GS15.** evaluate the information gathered from the customer report/ job card and assess repairs

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in repair & replacement of vehicle body and chassis components</i>	20	80	-	-
PC1. . assist the senior body technician to assess the overall damage to vehicle body and chassis components	1	3	-	-
PC2.. follow supervisors instructions as to which parts to restore or replace with respect to the auto body/ chassis components	1	3	-	-
PC3. . select, calibrate and use the appropriate tools and equipment for the body component/ chassis repair of the vehicle	1	5	-	-
PC4. . remove upholstery, accessories, electrical window-and-seat-operating equipment, and trim to gain access to vehicle bodies and fenders	1	5	-	-
PC5. . cut and tape plastic separating film to outside repair areas to avoid damaging surrounding surfaces during repair procedure, and remove tape and wash surfaces after repairs are complete	1	5	-	-
PC6. . assist in performing repair work under supervision of senior body technician such as: body panel repair minor structural damage major welded panels body components using lead wiping major sectional repair chassis/ frame and associated components	1	5	-	-
PC7. . assist in removal and replacement related work under supervision of Senior Body Technician such as: vehicle body panels, panel sections and ancillary fittings protector mouldings, transfers and decals mechanical units/assemblies electrical/electronic units/assemblies	1	4	-	-
PC8.. install various vehicle component seals	1	5	-	-
PC9. . prepare surface and apply window tainting	1	5	-	-
PC10. . file, grind, sand and smooth fill or repair surfaces, using power tools and hand tools	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11.. assist in clamping frames and sections to alignment machines that use hydraulic pressure to align damaged components	1	4	-	-
PC12. . assist in removal of damaged sections of vehicles using metal-cutting guns, air grinders and wrenches, and install replacement parts using wrenches or welding equipment	1	4	-	-
PC13. . assist in mixing polyester resins and hardeners to restore damaged areas	1	4	-	-
PC14. . assist in fitting and securing windows, vinyl roofs, and metal trim to vehicle bodies, using caulking guns, adhesive brushes, and mallets	1	4	-	-
PC15. . assist in filling small dents that cannot be worked out with plastic or solder	1	4	-	-
PC16. . assist in removal of small pits and dimples in body metal using pick hammers and punches	1	4	-	-
PC17. . ensure any malfunctions or repair requirements observed in vehicles (and beyond own scope of work) are reported to the concerned person	1	3	-	-
PC18. . ensure any malfunctions observed in tools and equipment are reported to the concerned person	1	3	-	-
PC19.. clean work areas, using air hoses, to remove damaged material and discarded fiberglass strips used in repair procedures	1	3	-	-
PC20.. ensure that trainings organized by the OEM from time-to-time are attended and knowledge levels are upgraded (esp. in case of newly launched products, product refreshes)	1	3	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1419
NOS Name	Assist in the repair and replacement of vehicle body and chassis components
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N9702: Carry out cleaning of a vehicle

Description

This OS unit is about carrying out routine daily washing and cleaning of the vehicle before the start of the trip as well as during trip

Elements and Performance Criteria

Washing of vehicle

To be competent, the user/individual on the job must be able to:

- PC1..** start the water pump and check for water supply in the pipe
- PC2..** thoroughly wash the external structure of the vehicle like the body, tyres, windshield , rear view mirrors etc. from the running water in pipe in order to clean the settled loose dust and dirt
- PC3..** while washing the vehicle with running water, to ensure that water does not enter into components like starter motor, alternator and horn.

Cleaning of vehicle

To be competent, the user/individual on the job must be able to:

- PC4..** rub the surfactant on the previously washed external structure properly using a spongy cloth in order to clean the hard-to-remove stains
- PC5..** ensure that the stains are not visible on the vehicle after rubbing.
- PC6..** while cleaning the vehicle, to observe and report any loose/broken/missing parts like mountings, seals, chassis fasteners

Re-washing and wiping

To be competent, the user/individual on the job must be able to:

- PC7..** after cleaning the vehicle, spray a jet of running water on the surfactant applied parts of vehicle in order to completely wash out the residue deposited on the vehicle.
- PC8..** while re - washing the vehicle with running water, to ensure that water does not enter into components like starter motor, alternator and horn.
- PC9..** take a clean wiping cloth and thoroughly wipe the whole vehicle including parts like dashboard , instrument panel , seats , steering wheel , driver console that cannot be surfactant cleaned as it would lead to malfunctioning

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** point of contact in case of unavailability of surfactant/cloth
- KU2.** any applicable rules and employee code of conduct stated by the organization
- KU3.** vehicle model and its appearance features which will aid him in cleaning
- KU4.** type of surfactant and cloth to be used for the cleaning process

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** execute task of cleaning and escalate and convey areas of concern to driver/supervisor
- GS2.** clean the various vehicle models in coordination with respective drivers
- GS3.** share requisite knowledge with newly joined cleaners
- GS4.** plan the cleaning activity so as to complete it in scheduled time

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Washing of vehicle</i>	6	22	-	-
PC1.. start the water pump and check for water supply in the pipe	1	7	-	-
PC2.. thoroughly wash the external structure of the vehicle like the body, tyres, windshield , rear view mirrors etc. from the running water in pipe in order to clean the settled loose dust and dirt	3	9	-	-
PC3.. while washing the vehicle with running water, to ensure that water does not enter into components like starter motor, alternator and horn.	2	6	-	-
<i>Cleaning of vehicle</i>	8	26	-	-
PC4.. rub the surfactant on the previously washed external structure properly using a spongy cloth in order to clean the hard-to-remove stains	3	9	-	-
PC5.. ensure that the stains are not visible on the vehicle after rubbing.	2	9	-	-
PC6.. while cleaning the vehicle, to observe and report any loose/broken/missing parts like mountings,seals, chassissfasteners	3	8	-	-
<i>Re-washing and wiping</i>	6	22	-	-
PC7.. after cleaning the vehicle, spray a jet of running water on the surfactant applied parts of vehicle in order to completely wash out the residue deposited on the vehicle.	2	8	-	-
PC8.. while re - washing the vehicle with running water, to ensure that water does not enter into components like starter motor, alternator and horn.	2	6	-	-
PC9.. take a clean wiping cloth and thoroughly wipe the whole vehicle including parts like dashboard , instrument panel , seats , steering wheel , driver console that cannot be surfactant cleaned as it would lead to malfunctioning	2	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	20	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9702
NOS Name	Carry out cleaning of a vehicle
Sector	Automotive
Sub-Sector	Road Transportation
Occupation	Driving
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ASC/N9708: Perform a routine check of basic functioning of vehicle and provide assistance to driver in servicing or minor repairs

Description

This OS unit is about carrying out basic technical checks on a vehicle

Elements and Performance Criteria

Basic technical check before the trip

To be competent, the user/individual on the job must be able to:

- PC1..** before start of the trip , refer a standard check list if available and ensure correct levels of engine oil coolant clutch, brake oil water in wiper washer
- PC2..** check all the tyres including spare tyre for inflation with a pressure gauge and ensure correct levels by deflating or inflating suitably
- PC3..** check that all lights, wiper and horn are working and report to the driver/supervisor in case of any malfunctioning
- PC4..** check the tool box and ensure availability of all the hand tools including jack,props,wheel choke etc
- PC5..** observe allexterior and interior parts of the vehicle for any damages to body panels, lights and wind shield , other glass parts and mirrors.

Escalation and assistance in case of technical problem

To be competent, the user/individual on the job must be able to:

- PC6..** in case of any problemobserved in any of the checkpoints, immediately inform the driver/supervisor
- PC7..** assist the driver in resolution of problem by providing him the required tools and equipments on the spot and physically lend him a helping hand in doing the strenuous jobs like lifting of vehicle by jack in case of punctured tyre replacement , opening of screws and fixtures for lights , wiper, tyres etc. while repairing and filling the engine/brake/coolant oil and other fluids to ensure optimum levels

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** point of contact to inform in case of any technical deficiencies
- KU2.** documents required for inspection as per government regulations (only familiarization with titles of various documents)
- KU3.** vehicle model and its various technical features
- KU4.** operating criteria for all the points mentioned in the checklist

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** technical standards of vehicle operation in terms of level of engine oil , illumination of various lights, air pressure in tyres etc.
- GS2.** follow driverinstructions about the rectification guidelines
- GS3.** how to learn from past mistakes and identify potential problems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic technical check before the trip</i>	15	57	-	-
PC1.. before start of the trip , refer a standard check list if available and ensure correct levels of engine oil coolant clutch, brake oil water in wiper washer	3	11	-	-
PC2.. check all the tyres including spare tyre for inflation with a pressure gauge and ensure correct levels by deflating or inflating suitably	3	12	-	-
PC3.. check that all lights, wiper and horn are working and report to the driver/supervisor in case of any malfunctioning	3	12	-	-
PC4.. check the tool box and ensure availability of all the hand tools including jack,props,wheel choke etc	3	11	-	-
PC5.. observe allexterior and interior parts of the vehicle for any damages to body panels, lights and wind shield , other glass parts and mirrors.	3	11	-	-
<i>Escalation and assistance in case of technical problem</i>	5	23	-	-
PC6.. in case of any problemobserved in any of the checkpoints, immediately inform the driver/supervisor	2	11	-	-
PC7.. assist the driver in resolution of problem by providing him the required tools and equipments on the spot and physically lend him a helping hand in doing the strenuous jobs like lifting of vehicle by jack in case of punctured tyre replacement , opening of screws and fixtures for lights , wiper, tyres etc. while repairing and filling the engine/brake/coolant oil and other fluids to ensure optimum levels	3	12	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9708
NOS Name	Perform a routine check of basic functioning of vehicle and provide assistance to driver in servicing or minor repairs
Sector	Automotive
Sub-Sector	Road Transportation
Occupation	Driving
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

ELE/N9905: Work effectively at the workplace

Description

This unit is about the communicating and managing work effectively at the workplace as well as taking measures to enhance own competence and working in a disciplined and ethical manner.

Scope

The scope covers the following :

- Communicate effectively at the workplace
- Work effectively
- Maintain and enhance professional competence
- Work in a disciplined and ethical manner
- Uphold social diversity at the workplace

Elements and Performance Criteria

Communicate effectively at the workplace

To be competent, the user/individual on the job must be able to:

- PC1.** exchange information and instruction with colleagues, and seek clarifications and feedback as necessary
- PC2.** assist colleagues where required
- PC3.** follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)
- PC4.** document and share all relevant information with stakeholders in agreed formats and as per agreed timelines

Work effectively

To be competent, the user/individual on the job must be able to:

- PC5.** identify and obtain clarity regarding organisational, team and own goals and targets
- PC6.** prioritise and plan work in order to achieve goals and targets
- PC7.** monitor own and team performance as per agreed plan
- PC8.** complete duties accurately, systematically and within required timeframes
- PC9.** express emotions appropriately at the workplace and manage own response to heightened emotions
- PC10.** maintain orderliness and cleanliness in the work area

Maintain and enhance professional competence

To be competent, the user/individual on the job must be able to:

- PC11.** identify own strengths and weaknesses in relation to goals and targets
- PC12.** adapt self, service, or product to meet success criteria
- PC13.** seek and select opportunities for continuous professional development
- PC14.** formulate a professional development plan to enhance capabilities

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- PC15.** build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations
- PC16.** examine developments and trends in field of work and their potential impact on work
- PC17.** take feedback from peers, supervisors and clients to improve own performance and practices

Work in a disciplined and ethical manner

To be competent, the user/individual on the job must be able to:

- PC18.** perform tasks as per workplace standards, organisational policies and legislative requirements
- PC19.** display appropriate professional appearance at the workplace and adhere to the organisational dress code
- PC20.** demonstrate responsible and disciplined behaviour at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behaviour at all times, adopting environment- friendly practices, etc.
- PC21.** identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution
- PC22.** protect the rights of the client and organisation when delivering services
- PC23.** ensure services are delivered equally to all clients regardless of personal and cultural beliefs
- PC24.** operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities
- PC25.** follow organisational guidelines and legal requirements on disclosure and confidentiality

Uphold social diversity at the workplace

To be competent, the user/individual on the job must be able to:

- PC26.** recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes
- PC27.** identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace
- PC28.** use inclusive or neutral language and gestures in all interactions
- PC29.** respect the personal and professional space of others
- PC30.** access grievance redressal mechanisms as per legislations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation's policies on dress code, workplace timings, workplace behaviour, performance management, incentives, delivery standards, information security, etc.
- KU2.** organizational hierarchy and escalation matrix
- KU3.** importance of the individual's role in the workflow
- KU4.** organisational norms on health, safety and sustainability
- KU5.** work area inspection procedures and practices
- KU6.** professional etiquette and grooming

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- KU7.** communication etiquette across communicative mediums (online, digital, and in-person) including strategies/methods for sharing information, documentation, and providing and receiving feedback
- KU8.** importance of self-evaluations and developing a continuous learning and professional development plan
- KU9.** developments and trends impacting professional practice
- KU10.** importance of taking and using feedback from colleagues and clients to identify and introduce improvements in work performance
- KU11.** professional ethics and workplace norms on reporting and/or penalizing unethical behaviour and practices.
- KU12.** guidelines and legal requirements on disclosure, confidentiality, and conflicts of interest
- KU13.** strategies for collaboration with colleagues and clients.
- KU14.** professional responses and strategies against inappropriate language or behaviour toward self and others
- KU15.** Implicit bias (based on gender, disability, class, caste, colour, race, culture, religion, etc.) and its consequences in the workplace
- KU16.** organizational guidelines, prevalent legislations and accessibility norms and processes to support PwDs at the workplace
- KU17.** strategies for time, effort and resource allocation towards the goals.
- KU18.** basic concepts of work productivity including waste reduction, efficient material usage and optimization of time

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation and forms such as work orders, invoices maintenance records activity logs, attendance sheets as per organizational format in English and/or local language
- GS2.** write basic accident or incident report accurately in an appropriate format
- GS3.** read warnings, instructions and other text material on product labels, components, etc. and relevant signages, warnings, labels or descriptions on equipment, etc. while carrying out work activities
- GS4.** convey and share technical information clearly using appropriate language
- GS5.** clarify task-related information
- GS6.** liaise with authorities and supervisors as per organizational protocol
- GS7.** listen, speak, and write in an inclusive, respectful manner in line with organizational protocol
- GS8.** seek clarification from immediate supervisor or responsible authority or exercise most appropriate solutions to safety breaches at work
- GS9.** report to the supervisor and when to deal with a colleague depending on the type of concern
- GS10.** deliver product to next work process on time
- GS11.** improve work process and report potential areas of delays and disruptions
- GS12.** communicate problems appropriately to others
- GS13.** identify symptoms of the fault to the cause of the problem and resolve, otherwise seek assistance and support from other sources to solve the problem

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- GS14.** anticipate and avoid hazards that may occur during repairs because of tools, materials used or repair processes
- GS15.** complete tasks efficiently and accurately within stipulated time
- GS16.** appreciate and respect social diversity in all professional settings
- GS17.** develop awareness and accountability for perspectives on gender, disabilities, and socio-cultural issues leading to discrimination, bias, or harassment at the workplace
- GS18.** maintain positive and effective relationships with colleagues and customers

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively at the workplace</i>	5	13	-	-
PC1. exchange information and instruction with colleagues, and seek clarifications and feedback as necessary	1	3	-	-
PC2. assist colleagues where required	1	3	-	-
PC3. follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)	1	4	-	-
PC4. document and share all relevant information with stakeholders in agreed formats and as per agreed timelines	2	3	-	-
<i>Work effectively</i>	6	13	-	-
PC5. identify and obtain clarity regarding organisational, team and own goals and targets	1	2	-	-
PC6. prioritise and plan work in order to achieve goals and targets	1	2	-	-
PC7. monitor own and team performance as per agreed plan	1	2	-	-
PC8. complete duties accurately, systematically and within required timeframes	1	2	-	-
PC9. express emotions appropriately at the workplace and manage own response to heightened emotions	1	2	-	-
PC10. maintain orderliness and cleanliness in the work area	1	3	-	-
<i>Maintain and enhance professional competence</i>	8	7	-	-
PC11. identify own strengths and weaknesses in relation to goals and targets	1	1	-	-
PC12. adapt self, service, or product to meet success criteria	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. seek and select opportunities for continuous professional development	1	1	-	-
PC14. formulate a professional development plan to enhance capabilities	2	1	-	-
PC15. build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations	1	1	-	-
PC16. examine developments and trends in field of work and their potential impact on work	1	1	-	-
PC17. take feedback from peers, supervisors and clients to improve own performance and practices	1	1	-	-
<i>Work in a disciplined and ethical manner</i>	11	16	-	-
PC18. perform tasks as per workplace standards, organisational policies and legislative requirements	2	2	-	-
PC19. display appropriate professional appearance at the workplace and adhere to the organisational dress code	1	2	-	-
PC20. demonstrate responsible and disciplined behaviour at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behaviour at all times, adopting environment- friendly practices, etc.	1	2	-	-
PC21. identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution	2	2	-	-
PC22. protect the rights of the client and organisation when delivering services	1	2	-	-
PC23. ensure services are delivered equally to all clients regardless of personal and cultural beliefs	1	2	-	-
PC24. operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. follow organisational guidelines and legal requirements on disclosure and confidentiality	1	2	-	-
<i>Uphold social diversity at the workplace</i>	10	11	-	-
PC26. recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes	2	2	-	-
PC27. identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace	2	2	-	-
PC28. use inclusive or neutral language and gestures in all interactions	2	2	-	-
PC29. respect the personal and professional space of others	2	2	-	-
PC30. access grievance redressal mechanisms as per legislations	2	3	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N9905
NOS Name	Work effectively at the workplace
Sector	Electronics
Sub-Sector	Generic
Occupation	Generic - Organizational Behaviour
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	24/02/2022
Next Review Date	30/12/2026
NSQC Clearance Date	30/12/2021

Qualification Pack

THC/N9916: Follow and Maintain Green Practices

Description

This unit is about adopting sustainable practices and optimizing use of resources, especially energy and waste, in day-to-day operations at work.

Scope

The scope covers the following :

- Following material conservation practices
- Ensuring effective waste management/recycling practices
- Ensuring use of eco-friendly practices

Elements and Performance Criteria

Following material conservation practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify ways to optimize usage of material including water in various tasks/activities
- PC2.** check for spills/leakages, plug them and escalate to appropriate authority if unable to rectify
- PC3.** ensure electrical equipment and appliances are switched off when not in use

Ensuring effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

- PC4.** identify recyclable and non-recyclable, and hazardous waste generated
- PC5.** dispose non-recyclable waste appropriately
- PC6.** follow processes specified for disposal of hazardous waste
- PC7.** ensure reuse and recycling of waste wherever applicable

Ensuring use of eco-friendly practices

To be competent, the user/individual on the job must be able to:

- PC8.** identify materials which can be replaced by environment friendly substitutes
- PC9.** follow SOPs to conserve and re-use water

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organization's procedures for minimizing waste
- KU2.** common sources of pollution and ways to minimize it
- KU3.** methods of optimum utilization of waste and best practices for waste disposal
- KU4.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** decide on appropriate waste disposal methods
- GS2.** analyse and select best suited environment friendly practices

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Following material conservation practices</i>	7	4	-	-
PC1. identify ways to optimize usage of material including water in various tasks/activities	4	1	-	-
PC2. check for spills/leakages, plug them and escalate to appropriate authority if unable to rectify	2	2	-	-
PC3. ensure electrical equipment and appliances are switched off when not in use	1	1	-	-
<i>Ensuring effective waste management/recycling practices</i>	13	5	-	-
PC4. identify recyclable and non-recyclable, and hazardous waste generated	4	1	-	-
PC5. dispose non-recyclable waste appropriately	4	2	-	-
PC6. follow processes specified for disposal of hazardous waste	1	1	-	-
PC7. ensure reuse and recycling of waste wherever applicable	4	1	-	-
<i>Ensuring use of eco-friendly practices</i>	8	2	-	-
PC8. identify materials which can be replaced by environment friendly substitutes	4	1	-	-
PC9. follow SOPs to conserve and re-use water	4	1	-	-
NOS Total	28	11	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	THC/N9916
NOS Name	Follow and Maintain Green Practices
Sector	Tourism & Hospitality
Sub-Sector	Tours and Travels
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	28/04/2022
Next Review Date	05/01/2026
NSQC Clearance Date	05/01/2023

Qualification Pack

HSS/N9624: Maintain a safe and secure working environment

Description

This OS unit is about the ensuring a safe and secure working environment

Scope

The scope covers the following :

- Workplace safety and security

Elements and Performance Criteria

Workplace safety and security

To be competent, the user/individual on the job must be able to:

- PC1.** identify potential hazards of safe work practices
- PC2.** use various hospital codes for emergency situations
- PC3.** comply with safety, and security procedures within the defined scope of competence and authority
- PC4.** provide Basic Life Support (BLS) and first aid whenever applicable under defined scope of work
- PC5.** follow organizations' procedures related to any emergency efficiently
- PC6.** report any identified breaches in health, safety, and security procedures to the designated person
- PC7.** complete any health and safety records accurately

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of health, safety, and security in the workplace
- KU2.** how to identify safety and security hazards
- KU3.** the importance of identifying individual responsibilities in relation to maintaining workplace safety and security requirements
- KU4.** the relevant up-to-date information on safety, and security that applies to the workplace
- KU5.** how to report any emergency
- KU6.** various hospital codes for emergency situations
- KU7.** how to create safety records and maintain them
- KU8.** concept of first aid and BLS
- KU9.** the importance of raising alarm about hazards for safety of others

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** read and understand organization policies and procedures
- GS2.** prepare status and progress reports
- GS3.** • communicate information (for example, facts, ideas, or messages) in a brief, clear, and
• organized manner
- GS4.** make decisions pertaining to the area of work
- GS5.** plan for safety of the work environment
- GS6.** identify hazards, evaluate possible solutions and suggest effective solutions
- GS7.** analyze the seriousness of hazards

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Workplace safety and security</i>	10	10	-	10
PC1. identify potential hazards of safe work practices	-	-	-	-
PC2. use various hospital codes for emergency situations	-	-	-	-
PC3. comply with safety, and security procedures within the defined scope of competence and authority	-	-	-	-
PC4. provide Basic Life Support (BLS) and first aid whenever applicable under defined scope of work	-	-	-	-
PC5. follow organizations' procedures related to any emergency efficiently	-	-	-	-
PC6. report any identified breaches in health, safety, and security procedures to the designated person	-	-	-	-
PC7. complete any health and safety records accurately	-	-	-	-
NOS Total	10	10	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	HSS/N9624
NOS Name	Maintain a safe and secure working environment
Sector	Healthcare
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	28/04/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AAS/N0502.Follow safety and security procedures	52	48	-	-	100	10
ASC/N0001.Plan and organise work to meet expected outcomes	25	75	-	-	100	8
ASC/N1404.Carry out diagnosis of vehicle for repair requirements	30	70	-	-	100	10
ASC/N1405.Carry out service and major repairs in mechanical aggregates and overhauling of a vehicle	30	70	-	-	100	12
ASC/N1406.Carry out service and repairs of electrical and electronic faults in a vehicle	25	75	-	-	100	12
ASC/N1417.Carry out painting work on vehicles and replace/install painted body panels	20	80	-	-	100	10
ASC/N1419.Assist in the repair and replacement of vehicle body and chassis components	20	80	-	-	100	10
ASC/N9702.Carry out cleaning of a vehicle	20	70	-	-	90	8

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9708.Perform a routine check of basic functioning of vehicle and provide assistance to driver in servicing or minor repairs	20	80	-	-	100	10
ELE/N9905.Work effectively at the workplace	40	60	-	-	100	5
THC/N9916.Follow and Maintain Green Practices	28	11	-	-	39	5
Total	310	719	-	-	1029	100

Optional: 1 Maintain a safe and secure working environment

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HSS/N9624.Maintain a safe and secure working environment	10	10	-	10	30	5
Total	10	10	0	10	30	5

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.